1. ss

Displays active network connections and listening ports.

```
vanam@Uday:~$ sudo ss -tulpn
Netid  State   Recv-Q  Send-Q    Local Address:Port      Peer Address:Port  Process
udp    UNCONN  0       0           127.0.0.54:53           0.0.0.0:*      users:(("systemd-resolve",pid=160,fd=16))
udp    UNCONN  0       0         127.0.0.53%lo:53          0.0.0.0:*      users:(("systemd-resolve",pid=160,fd=14))
udp    UNCONN  0       0        10.255.255.254:53          0.0.0.0:*
udp    UNCONN  0       0            127.0.0.1:323          0.0.0.0:*
udp    UNCONN  0       0                [::1]:323             [::]:*
tcp    LISTEN  0       1000     10.255.255.254:53          0.0.0.0:*
tcp    LISTEN  0       4096      127.0.0.53%lo:53          0.0.0.0:*      users:(("systemd-resolve",pid=160,fd=15))
tcp    LISTEN  0       4096          127.0.0.1:32837       0.0.0.0:*      users:(("containerd",pid=252,fd=15))
tcp    LISTEN  0       4096         127.0.0.54:53          0.0.0.0:*      users:(("systemd-resolve",pid=160,fd=17))
tcp    LISTEN  0       50                 *:8080                *:*      users:(("java",pid=179,fd=9))
```

- `-t` – TCP.
- `-u` – UDP.
- `-l` – Listening ports.
- `-p` – Process information.
- `-n` – Numeric addresses and ports.

2. tracepath

Identifies the network path packets take to reach a destination.

```
vanam@Uday:~$ tracepath google.com
 1?: [LOCALHOST]                      pmtu 1500
 1:  Uday.mshome.net                                       0.714ms
 1:  Uday.mshome.net                                       0.549ms
 2:  reliance.reliance                                     2.635ms
 3:  10.48.160.1                                           5.212ms
 4:  172.31.0.144                                          5.985ms
 5:  192.168.68.142                                        5.895ms asymm  6
 6:  172.26.77.164                                         4.249ms asymm  7
 7:  172.26.77.131                                         5.955ms asymm  9
 8:  192.168.68.134                                        6.738ms asymm 10
```

3. ip neigh

Displays the system's IP-to-MAC address neighbor table.

```
vanam@Uday:~$ ip neigh show
172.19.176.1 dev eth0 lladdr 00:15:5d:7b:46:0b REACHABLE
vanam@Uday:~$
```

4. ethtool

Displays network interface information.

```
vanam@Uday:~$ sudo ethtool eth0
Settings for eth0:
        Supported ports: [  ]
        Supported link modes:   Not reported
        Supported pause frame use: No
        Supports auto-negotiation: No
        Supported FEC modes: Not reported
        Advertised link modes:  Not reported
        Advertised pause frame use: No
        Advertised auto-negotiation: No
        Advertised FEC modes: Not reported
        Speed: 10000Mb/s
        Duplex: Full
        Auto-negotiation: off
        Port: Other
        PHYAD: 0
        Transceiver: internal
        Current message level: 0x000000f7 (247)
                               drv probe link ifdown ifup rx_err tx_err
        Link detected: yes
```

5. timeout

Stops a command if it runs longer than a specified duration.

```
vanam@Uday:~$ timeout 10s ping google.com
PING google.com (142.251.43.206) 56(84) bytes of data.
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=1 ttl=111 time=7.52 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=2 ttl=111 time=16.2 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=3 ttl=111 time=7.67 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=4 ttl=111 time=10.5 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=5 ttl=111 time=7.27 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=6 ttl=111 time=9.39 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=7 ttl=111 time=8.41 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=8 ttl=111 time=9.36 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=9 ttl=111 time=8.33 ms
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=10 ttl=111 time=9.64 ms
vanam@Uday:~$
```

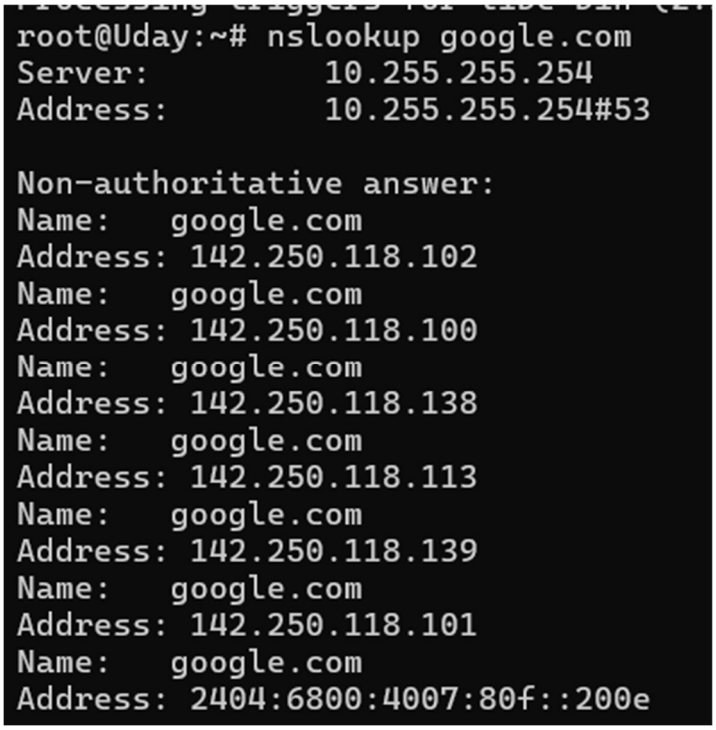
6. systemd-analyze

Analyzes Linux boot performance.

```
vanam@Uday:~$ systemd-analyze blame
14.305s apt-daily.service
12.616s jenkins.service
 2.363s docker.service
 1.915s landscape-client.service
  596ms containerd.service
  413ms snapd.seeded.service
  399ms snapd.service
  375ms dev-sdd.device
  215ms docker.socket
  199ms systemd-journal-flush.service
  198ms systemd-resolved.service
  194ms user@1000.service
  167ms logrotate.service
  129ms systemd-udev-trigger.service
  127ms packagekit.service
  124ms rsyslog.service
  105ms systemd-timesyncd.service
   98ms systemd-udevd.service
   66ms systemd-logind.service
   60ms dpkg-db-backup.service
   60ms systemd-tmpfiles-setup.service
   60ms keyboard-setup.service
   58ms e2scrub_reap.service
```

7.nslookup

Checks DNS resolution.



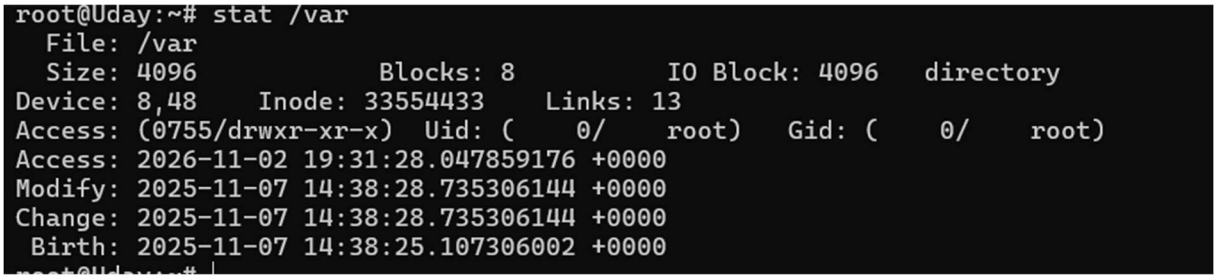
8. tcpdump

Captures and analyzes network packets.



9. stat

Displays detailed information about a file or directory.



10. watch

Repeatedly executes a command and displays updated output.

